FOR THE TUTOR OR PARENT

Answer key: Ages 11-13

Accept equivalent forms (for example 2.05 and $2.05). One mark per question. Keep this key away from the child until they have finished.

Warm-up (4 marks)

#	Question	Answer	Watch for
1	Write every factor of 24 (every whole number that divides 24 exactly).	**1, 2, 3, 4, 6, 8, 12, 24**	All eight needed for the mark. Common error: missing 8 or 12, or stopping at 6. Skill used in Lessons 2 and 8.
2	Write 6/36 in its simplest form.	**1/6**	Divide top and bottom by 6. Common error: 3/18 or 2/12 (not fully simplified). Skill used in Lessons 6 and 9.
3	Two angles of a triangle are 52 and 63 degrees. What is the third angle, in degrees?	**65**	Angle sum is 180. Common error: using 360 or an arithmetic slip. Skill used in Lesson 7.
4	Read this little program. Start with n = 5. Then repeat 4 times: change n by 3 (add 3 to n). At the end, say n. What does it say?	**17**	5, 8, 11, 14, 17. Common error: 20 (adding 3 five times) or 12 (adding 3 only once). Not a mark against the child: this also shows how ready they are for the coding side, so the tutor can plan support.

Main questions (6 marks)

#	Question	Answer	Watch for
5	A number machine doubles the input and then adds 5. The output is 31. What was the input?	**13**	Look for working backwards (31 - 5, then halve). Common error: halving first or giving 26.
6	A prime number has exactly two factors: 1 and itself. For example, 7 is prime, but 9 is not (its factors are 1, 3 and 9). How many prime numbers are there between 30 and 50?	**5**	Primes are 31, 37, 41, 43, 47. Common errors: including 39 or 49, or counting 1.
7	The temperature is -4 degrees C at night. It rises by 9 degrees by noon, then falls by 12 degrees by evening. What is the evening temperature in degrees C?	**-7**	Check number line use: -4 + 9 = 5, 5 - 12 = -7. Common error: answering 7.
8	A bag has 4 red, 3 blue and 5 green marbles. One marble is picked without looking. The chance (probability) of picking blue is 3 out of 12, written 3/12. What is the probability that it is NOT green? Give a fraction in simplest form.	**7/12**	Total is 12, not green is 7. Common error: 5/12, or 7/5 (comparing to green).
9	The average of 5 numbers is 12. Four of the numbers are 8, 10, 15 and 11. What is the fifth number?	**16**	Total must be 60 (5 x 12); the four add to 44. Common error: finding the average of the four numbers. Some children say 'mean' for average; both are fine.
10	It is 10 o'clock now on a 12-hour clock. What time will it be 17 hours later? Give the hour only.	**3**	10 + 17 = 27, 27 - 24 = 3 (or 17 mod 12 = 5, then 10 + 5 = 15 -> 3). Common error: answering 27 or 5.

Brain stretchers (2 marks, optional)

#	Question	Answer	Watch for
11	The pattern 4, 7, 10, 13, ... carries on in the same way. What is the 50th number? Explain how you worked it out.	**151**	Rule: 4 + 3 x 49 = 151 (the 50th number has 49 jumps of 3). Accept any clear method. Common error: 150 (3 x 50) or 154 (4 + 3 x 50). A flag for the Think harder pages, not a mark against the child.

12	Using the digits 1, 2, 3 and 4, with no digit used more than once, how many different 3-digit even numbers can be made?	**12**	Last digit 2 or 4 (2 ways), then 3 x 2 for the others. Common error: 24 (ignoring evenness) or listing incompletely. A flag for the Think harder pages, not a mark against the child.

Questions the child ticked "not learnt yet" score 0 but tell you what to teach first. They do not count against the child.